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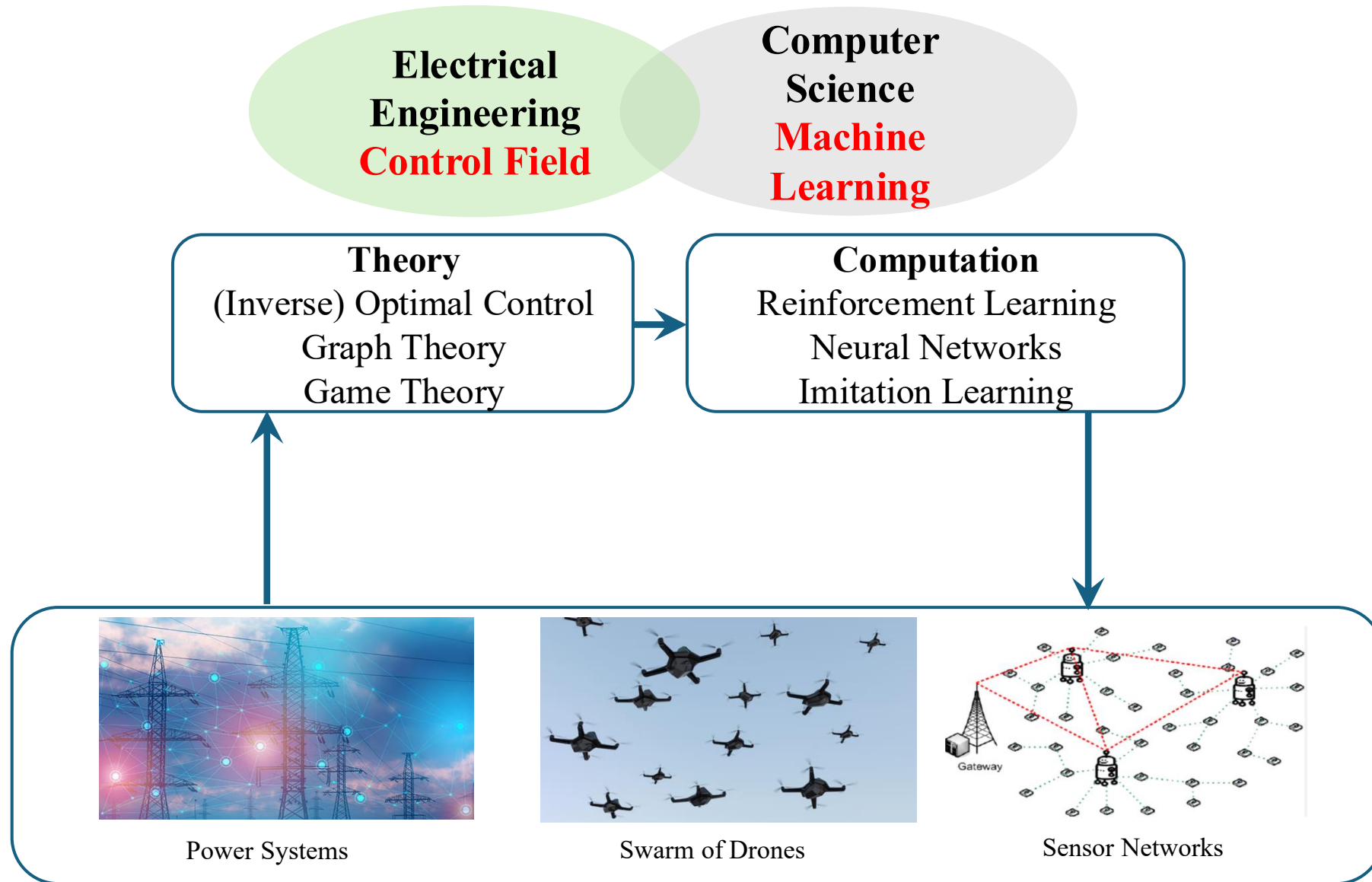
Invited by Prof. Ya Zhang

Cooperative Control with Integral and Inverse Reinforcement Learning

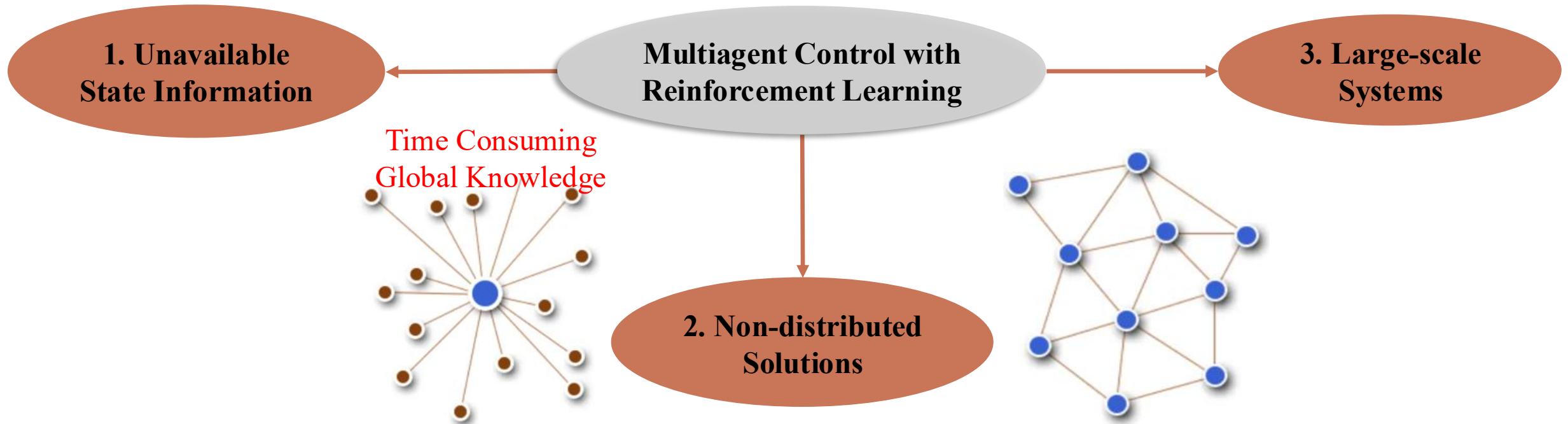
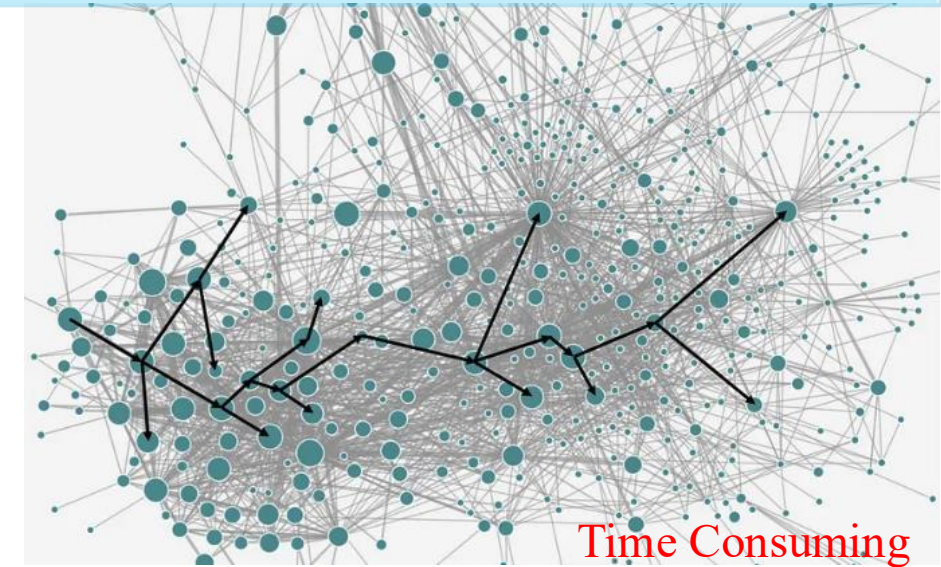
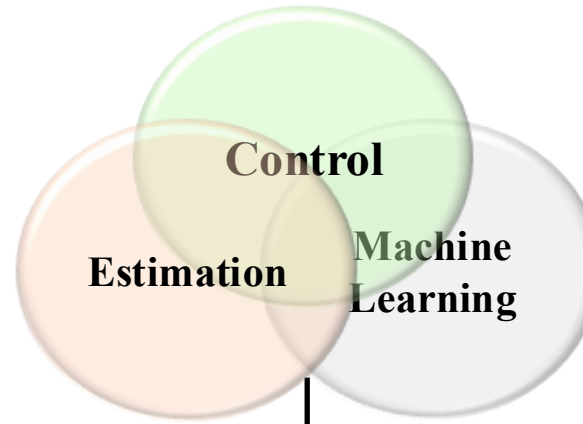
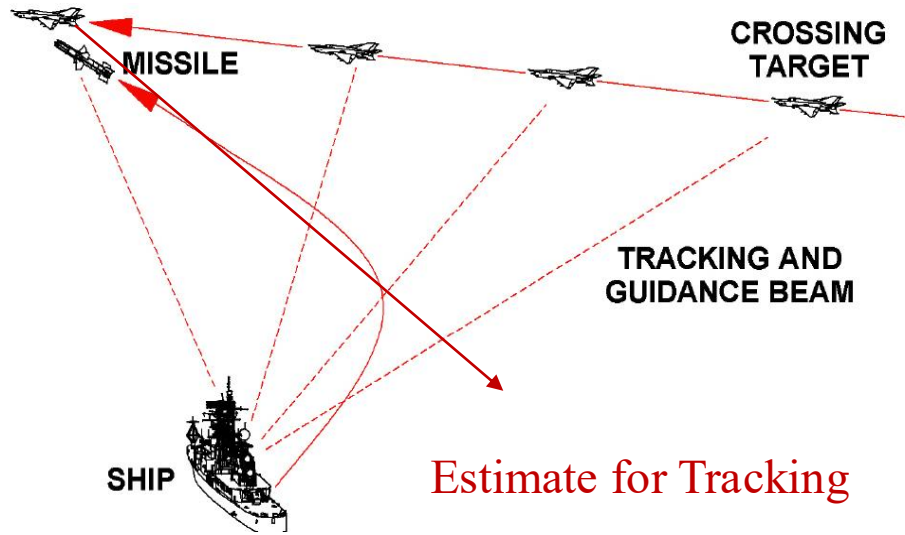
Dr. Bosen Lian, PhD, Assistant Professor, Auburn University



Intelligent Cooperative Control



Challenges in Cooperative Control with Reinforcement Learning



To Address the Challenges

Distributed Estimation-based Tracking Control with Reinforcement Learning



**1. Unavailable
State Information**

Distributed Control for Multiplayer Games with Reinforcement Learning



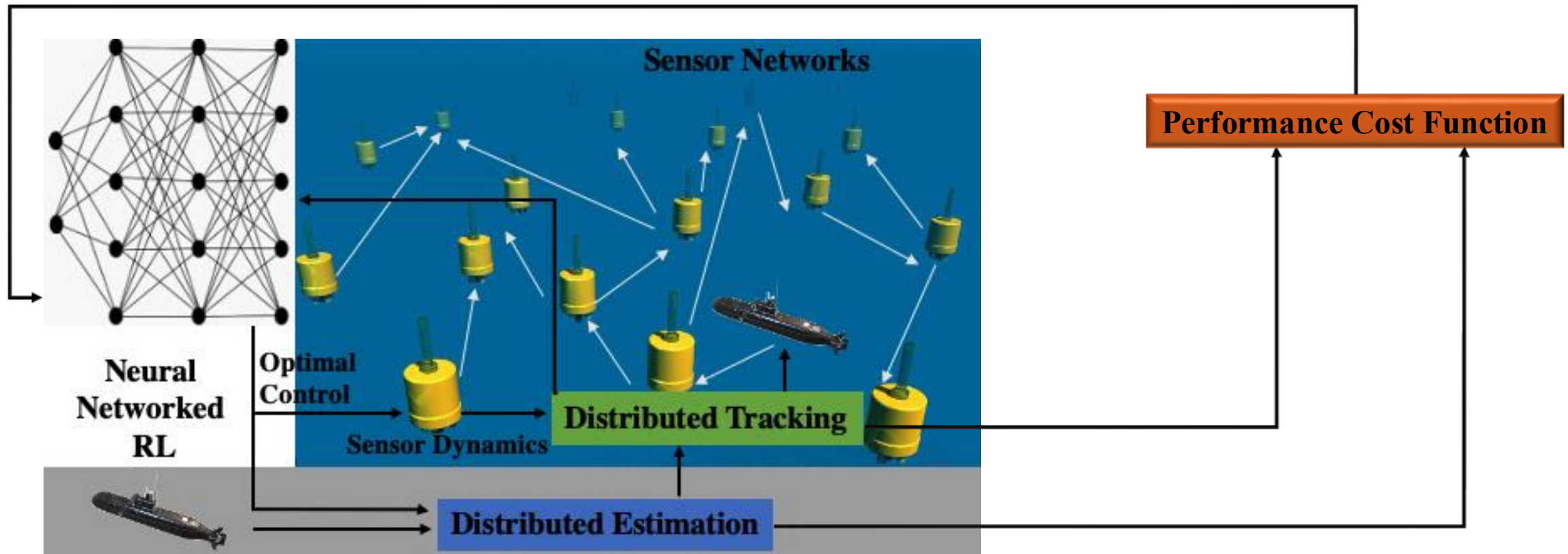
**2. Distributed
Solutions**

Accelerated Reinforcement Learning for Large-scale Multiagent Systems

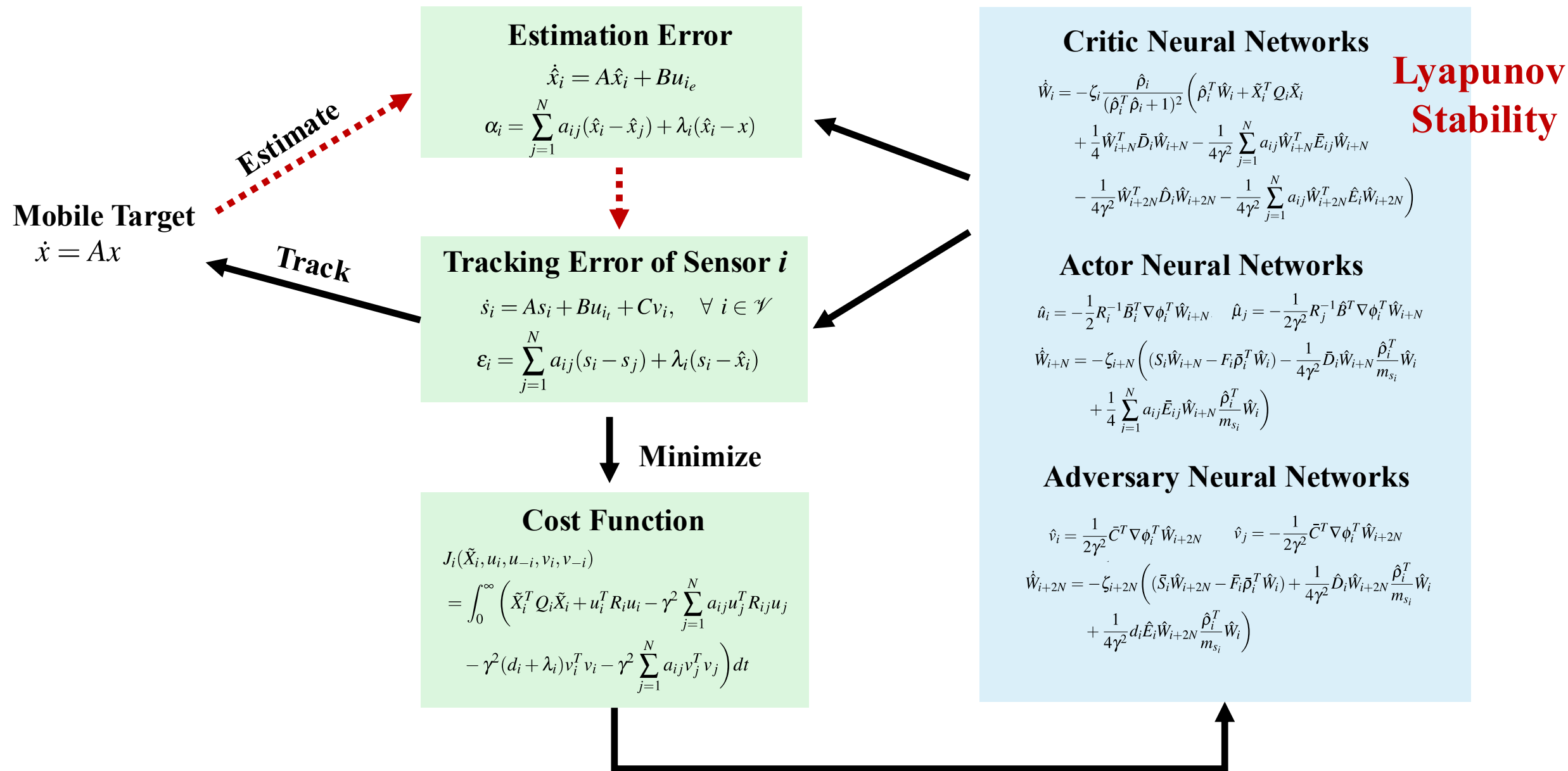


**3. Scale-free
Systems**

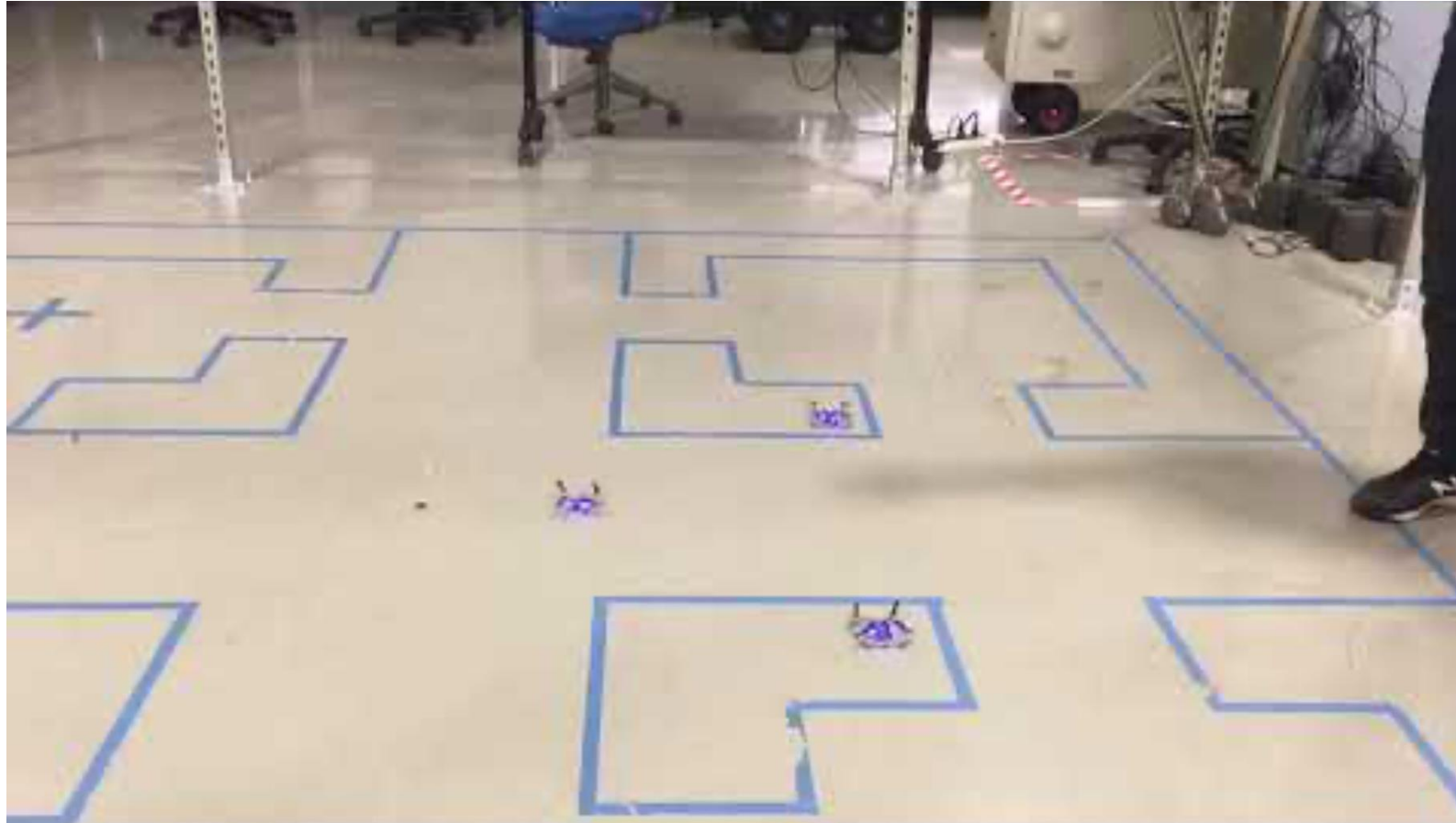
Distributed Estimation Based Tracking Control of Multiagent Sensor Networks with Reinforcement Learning



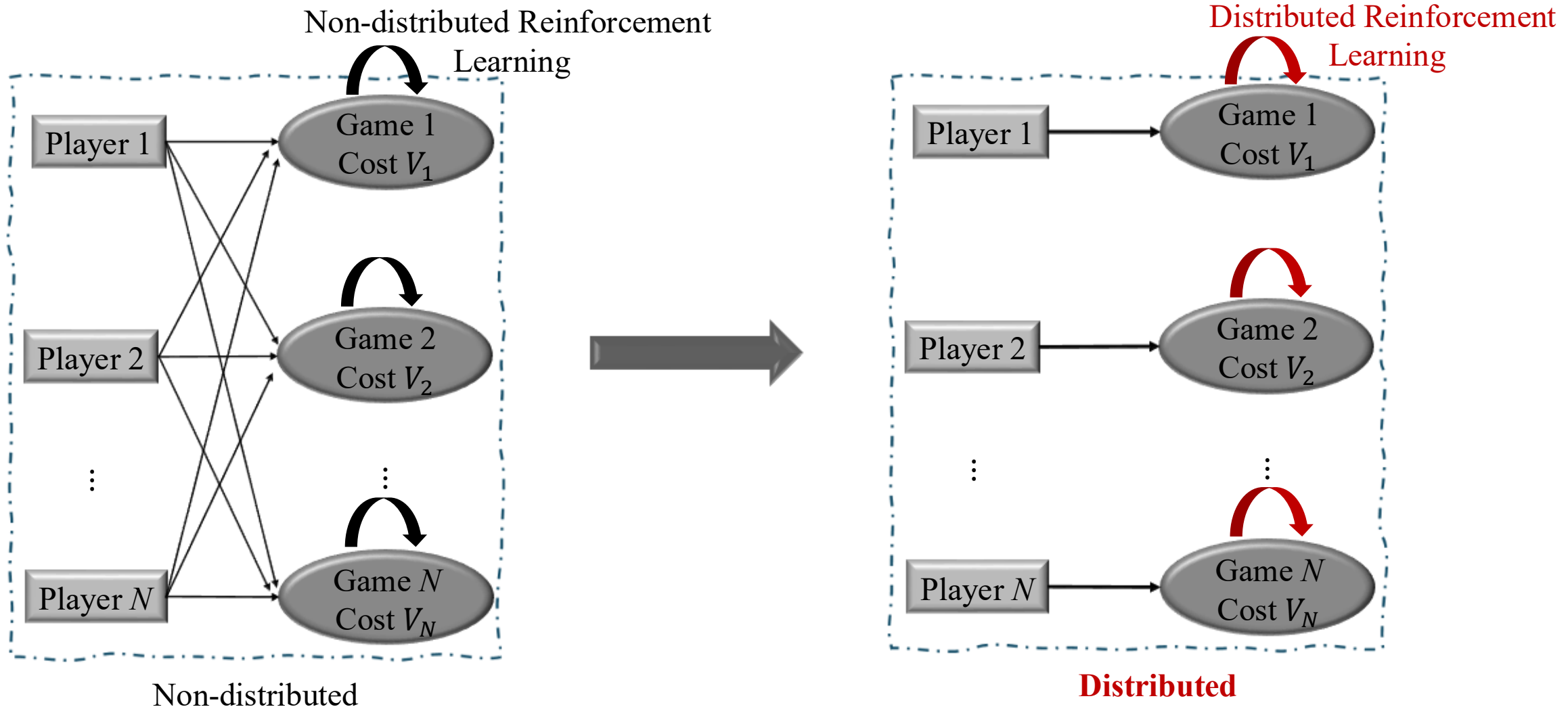
Realtime Distributed Adversary-Actor-Critic Neural Networks



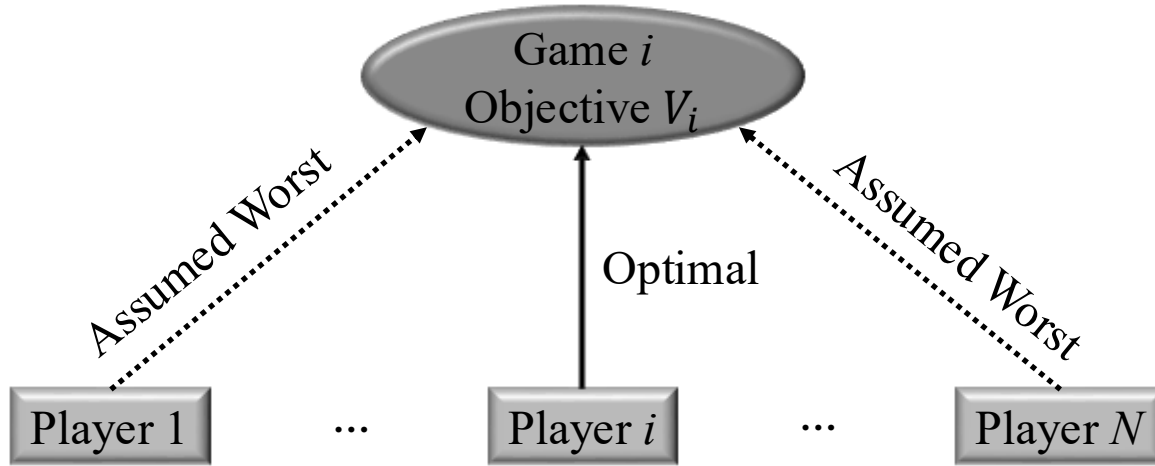
3 Sensors Track the Target



Distributed Control for Multiplayer Games with Reinforcement Learning



Distributed Model-free Reinforcement Learning



Distributed Minimax Control Strategy

Algorithm 2 Data-driven Off-policy RL Algorithm to solve the Distributed Minmax Strategies in Multiplayer Games

- 1) Select initial stabilizing control policies K_i^0 and $K_{ij}^0 = 0$ for $i \in \mathcal{N}$, $j \in -i$. Set small thresholds e_i and $h = 0$. Apply $u_i = -K_i x + \varepsilon_i$ to (51), where K_i is a fixed stabilizing policy and ε_i is a small probing noise.
- 2) **Solve control feedback gains K_i^{h+1} and K_{ij}^{h+1} by**

$$\begin{aligned} & x^T(t + \Delta t) P_i^h x(t + \Delta t) - x^T(t) P_i^h x(t) \\ & - \int_t^{t+\Delta t} 2(u_i - u_i^h)^T R_{ii} K_i^{h+1} x d\tau \\ & + \int_t^{t+\Delta t} 2\gamma_i^2 \sum_{j \in -i} (u_j - v_{ij}^h)^T R_{ij} K_{ij}^{h+1} x d\tau \\ & = - \int_t^{t+\Delta t} \left(x^T Q_i x + (u_i^h)^T R_{ii} u_i^h - \gamma_i^2 \sum_{j \in -i} (v_{ij}^h)^T R_{ij} v_{ij}^h \right) d\tau. \end{aligned}$$

- 3) **Update control inputs u_i^{h+1} and v_{ij}^{h+1} by**

$$\begin{aligned} u_i^{h+1} &= -K_i^{h+1} x, \\ v_{ij}^{h+1} &= -K_{ij}^{h+1} x. \end{aligned}$$

- 4) **Stop if $\|P_i^{h+1} - P_i^h\| \leq e_i$, $\forall i \in \mathcal{N}$. Otherwise, set $h \rightarrow h + 1$, and go to Step 2.**

Asymptotic Stability

Distributed minimax strategy

$$u_i^* \triangleq \arg \min_{u_i} \max_{u_{-i}} V_i(x, u_i, u_{-i})$$



Lyapunov Theory

Multiplayer dynamic systems

$$\dot{x} = \left(A - \sum_{j=1}^N B_j R_{jj}^{-1} B_j^T P_j \right) x$$

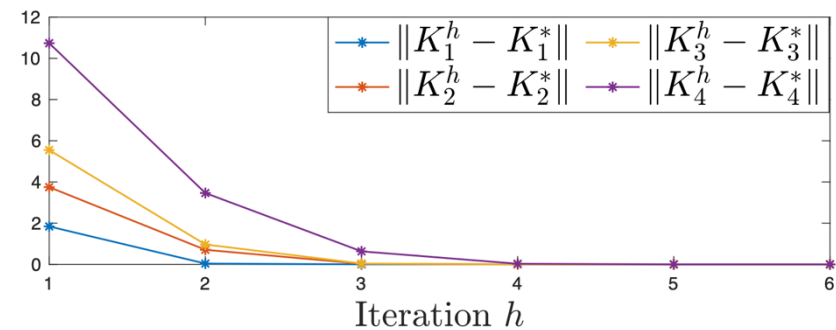
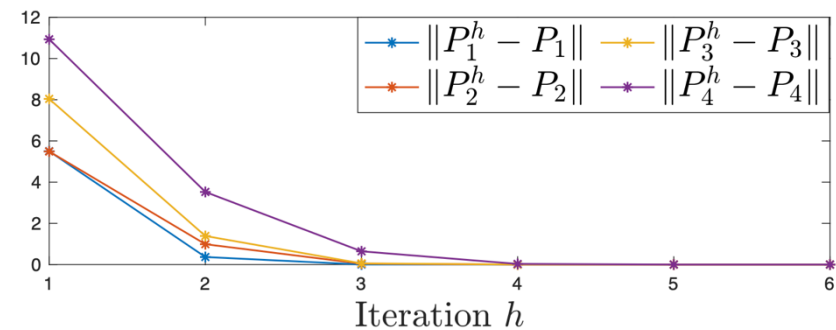
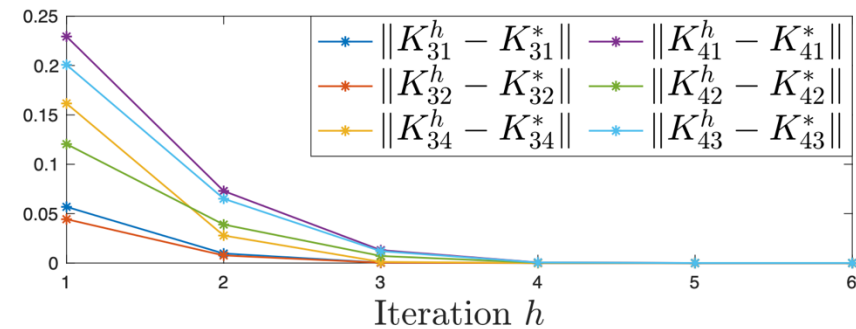
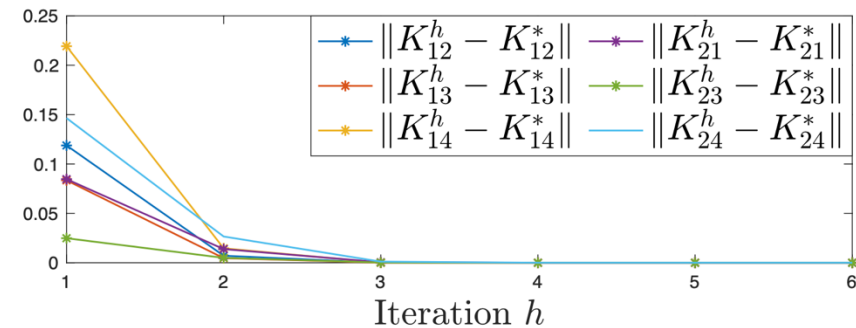
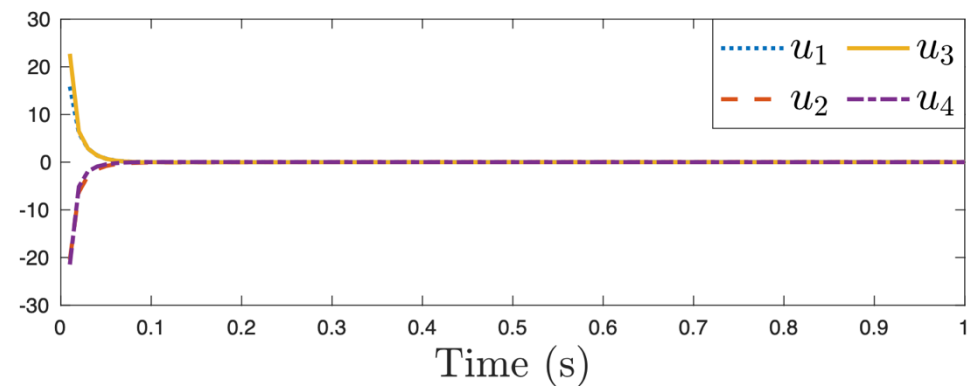
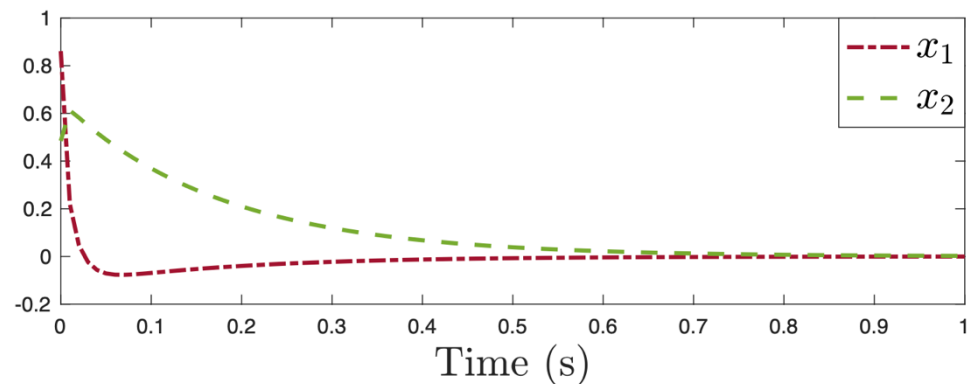
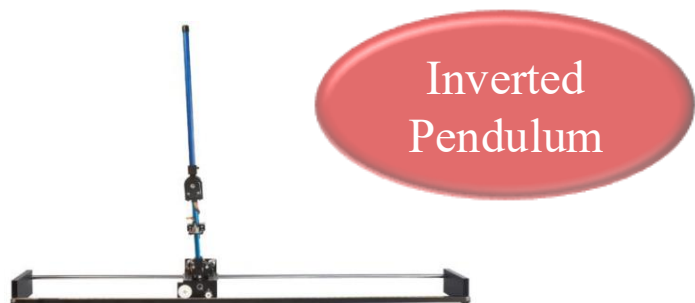


$$\lim_{t \rightarrow \infty} x(t) = 0$$

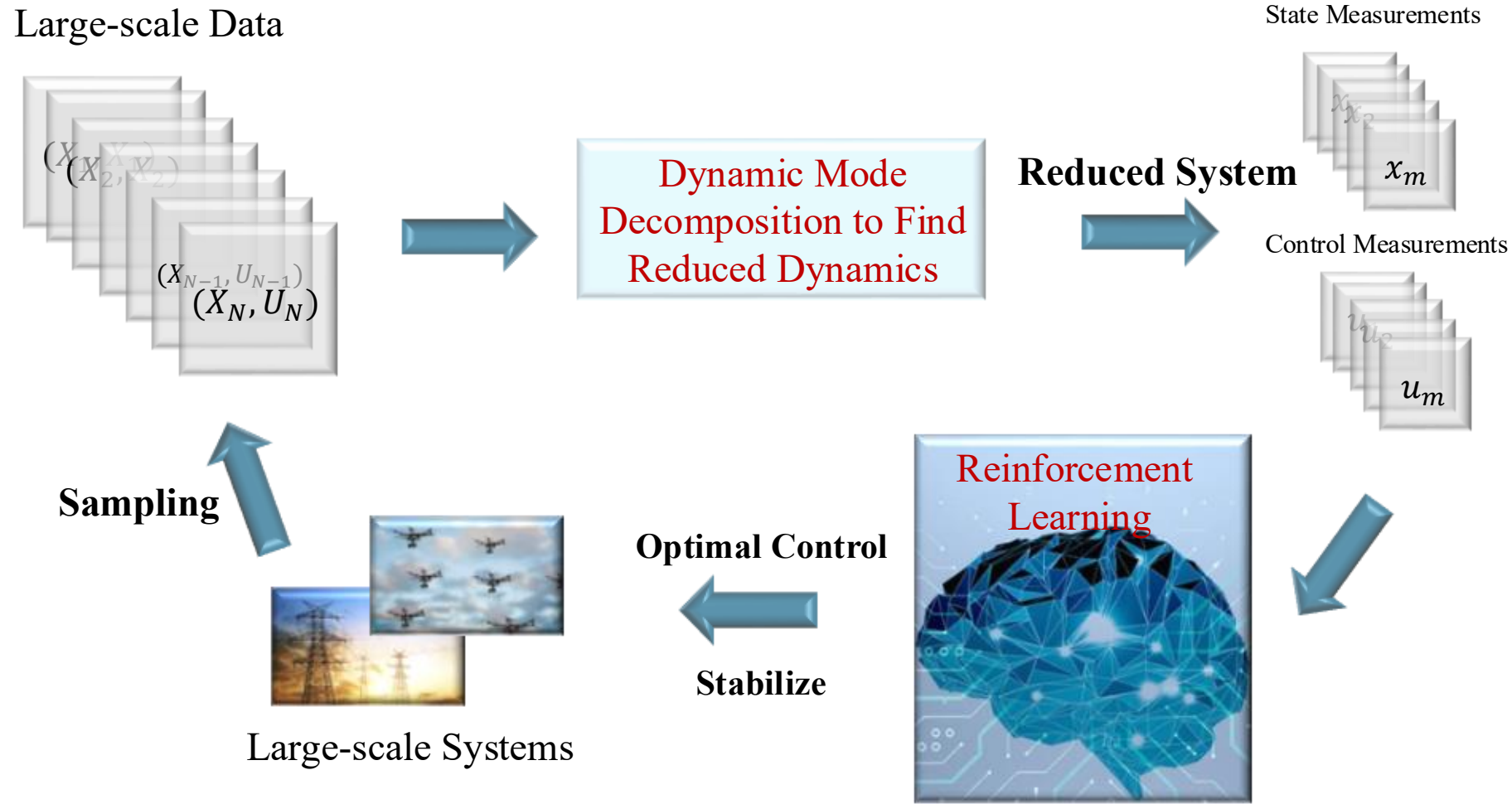
Improved Robustness

- Phase margin: $|PM| < 60^\circ + \omega_i$
- Gain margin: $\frac{a}{2} < GM < \infty$

where $\omega_i > 0$ and $0 < a < 1$.



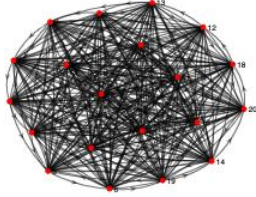
Accelerated Reinforcement Learning for Large-Scale Multiagent Systems



Low-Scale Subsystems

$$x_{i_k} \in \mathbb{R}^n, u_{i_k} \in \mathbb{R}^m$$

$$x_{i_{k+1}} = A_{ii}x_{i_k} + A_{ij} \sum_{j \neq i, j=1}^M e_{ij}(x_{i_k} - x_{j_k}) + B_i u_{i_k}, \quad i \in \mathcal{M}$$



M agents

Large-Scale Global Systems

$$\tilde{x}_k \in \mathbb{R}^{nM}, \tilde{u}_k \in \mathbb{R}^{mM}$$

$$\tilde{x}_{k+1} = \tilde{A}\tilde{x}_k + \tilde{B}\tilde{u}_k$$

$$J(\tilde{u}_k, \tilde{x}_k) = \sum_{k=0}^{\infty} [\tilde{x}_k^T \tilde{Q} \tilde{x}_k + \tilde{u}_k^T \tilde{R} \tilde{u}_k]$$

Reduced Scale Global Systems

$$\tilde{\xi}_k \in \mathbb{R}^r$$

$$\xi_{k+1} = Y\tilde{A}Y^\dagger \xi_k + Y\tilde{B}\tilde{u}_k$$

$$\bar{J}(\tilde{u}_k, \xi_k) = \sum_{k=0}^{\infty} [\xi_k^T \tilde{Q}_r \xi_k + \tilde{u}_k^T R \tilde{u}_k]$$

Dynamic Mode Decomposition

$$\xi_k \approx Y\tilde{x}_k$$

$$\theta = \frac{1}{\lambda} X_2 \tilde{V} \tilde{S}^{-1} v$$

$$\tilde{T}v = X_2 \tilde{V} \tilde{S}^{-1} \tilde{U}^T v = YX_2 \tilde{V} \tilde{S}^{-1} v = \tilde{T}v = \lambda v.$$

$$X_1 = \begin{bmatrix} | & | & \dots & | \\ x_1 & x_2 & \dots & x_{p-1} \\ | & | & \dots & | \end{bmatrix}, \quad X_2 = \begin{bmatrix} | & | & \dots & | \\ x_2 & x_3 & \dots & x_p \\ | & | & \dots & | \end{bmatrix}$$

$$r \ll nM$$

Sampling

Algorithm 2 Dynamic Mode Decomposition-based Model-free Reinforcement Learning for Large-scale Systems

- 1: **Initialization:** Given N , set $j = 0$ and small threshold ϵ . Select stabilizing $\tilde{K}_r^0, \tilde{K}^0$, and control input $\tilde{u}_k = -\tilde{K}\tilde{x}_k + \epsilon_k$, where ϵ_k is a probing noise.
- 2: **Data Collection:** Same as step 2 in Algorithm 1 and compute (20).
- 3: **DMD Preconditioning:** Set strict threshold μ . Compute reduced-order data vectors $(\hat{\rho}_{xx}, \hat{\rho}_{xu}, \hat{\rho}_{uu})$ in (41) and operators (ϕ_r^j, ψ^j) given by

$$\phi_r^j = \begin{bmatrix} \xi_k^T \tilde{Q}_r \xi_k + \xi_k^T (\tilde{K}_r^j)^T \tilde{R} \tilde{K}_r^j \xi_k \\ \vdots \\ \xi_{k+\zeta-1}^T \tilde{Q}_r \xi_{k+\zeta-1} + \xi_{k+\zeta-1}^T (\tilde{K}_r^j)^T \tilde{R} \tilde{K}_r^j \xi_{k+\zeta-1} \end{bmatrix},$$

$$\psi^j = [\hat{\rho}_{xx} \quad \hat{\rho}_{xu} \quad \hat{\rho}_{uu}],$$

$$\hat{\rho}_{xx} = \rho_{xxs}(\theta \otimes \theta) \in \mathbb{R}^{1 \times r^2},$$

$$\hat{\rho}_{xu} = \rho_{xus}(I_{mM} \otimes \theta) \in \mathbb{R}^{1 \times r(mM)},$$

$$\hat{\rho}_{uu} = \rho_{uus}(I_{mM} \otimes I_{mM}) \in \mathbb{R}^{1 \times (mM)^2}.$$

- 4: **Policy Evaluation:** Compute $\tilde{P}_r^{j+1}$ by

$$((\psi_r^j)^T \psi_r^j)^{-1} (\psi_r^j)^T \phi_r^j = \begin{bmatrix} \text{vec}(\tilde{P}_r^{j+1})^T \\ \text{vec}((Y\tilde{B})^T \tilde{P}_r^{j+1} Y\tilde{A}Y^\dagger)^T \\ \text{vec}((Y\tilde{B})^T \tilde{P}_r^{j+1} Y\tilde{B})^T \end{bmatrix}.$$

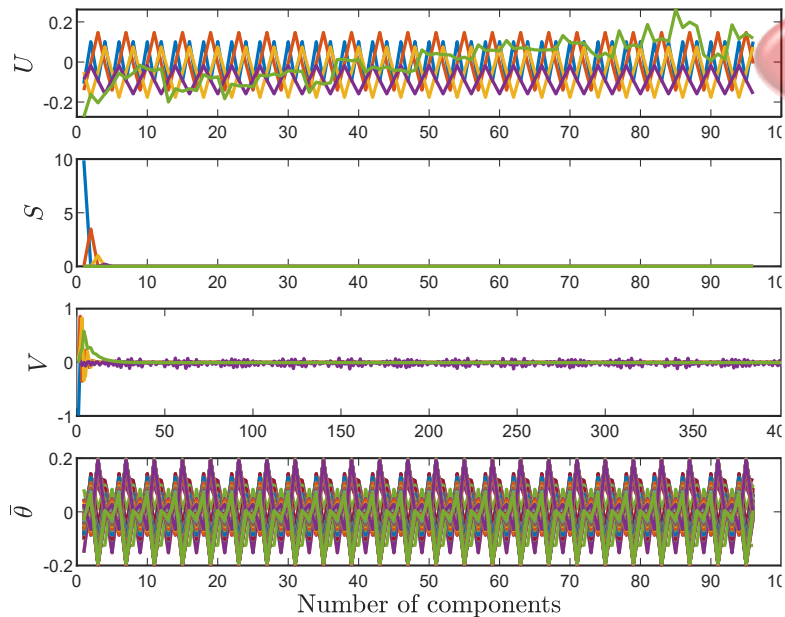
- 5: **Policy Improvement:** Compute $\tilde{K}_r^{j+1}$ by

$$\tilde{K}_r^{j+1} = (R + (Y\tilde{B})^T \tilde{P}_r^{j+1} Y\tilde{B})^{-1} (Y\tilde{B})^T \tilde{P}_r^{j+1} A.$$

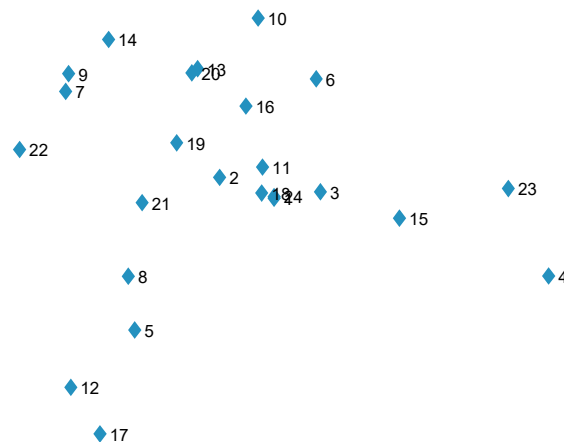
- 6: **Stop** if $\|\tilde{K}_r^{j+1} - \tilde{K}_r^j\| \leq \epsilon$. Otherwise, set $j = j + 1$, and go to step 2.

- 7: **Collect:**

$$\tilde{K}^{j+1} = Y\tilde{K}_r^{j+1}.$$

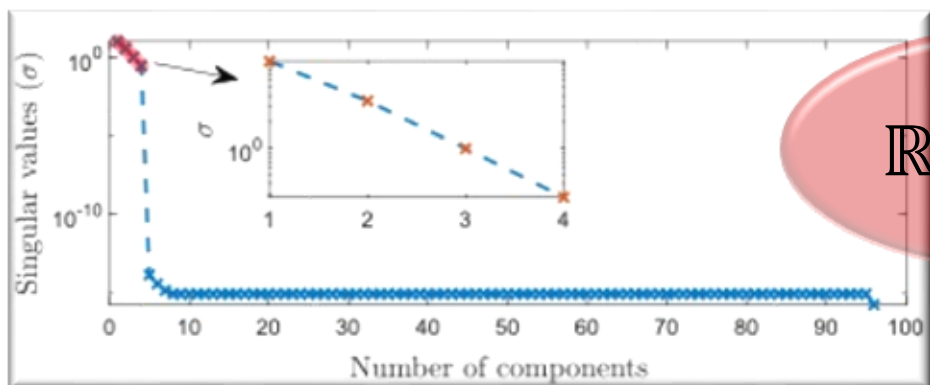


$\mathbb{R}^{96}$



Total dynamic modes of 24-area power system

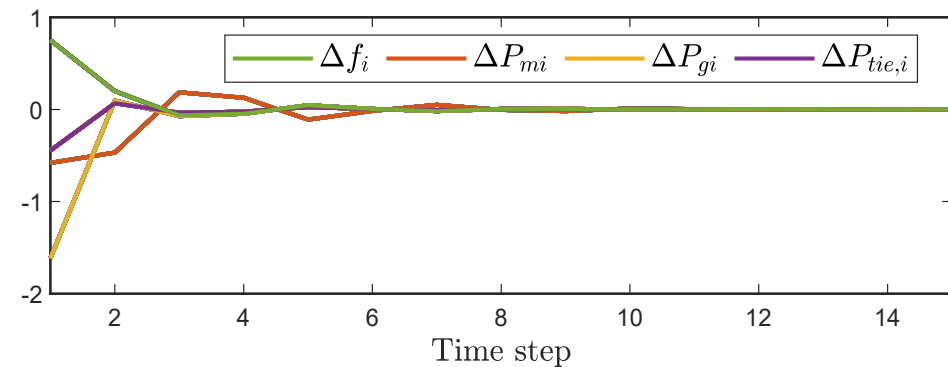
Dynamic Mode Decomposition



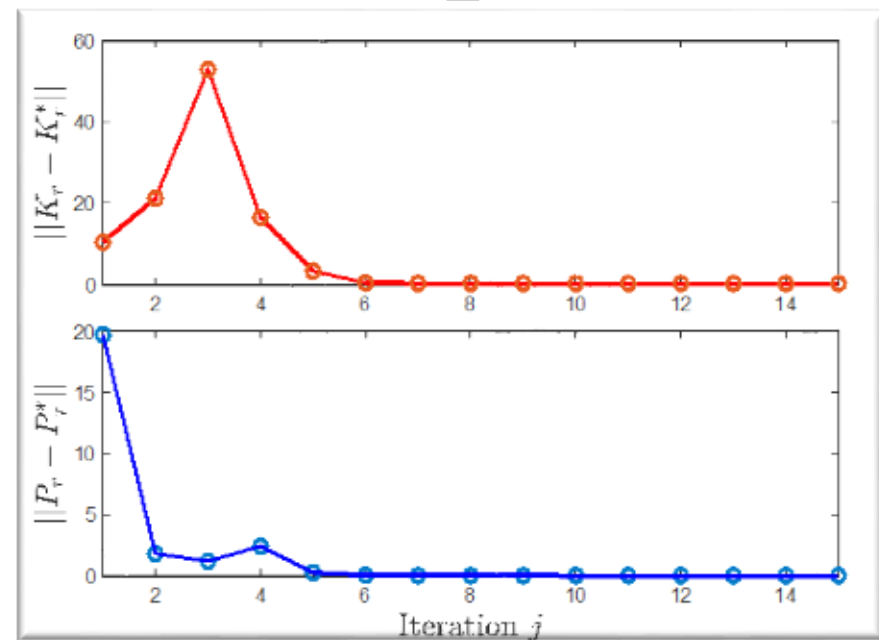
4 dominant singular values

$\mathbb{R}^{96}$ to $\mathbb{R}^4$

Reinforcement Learning



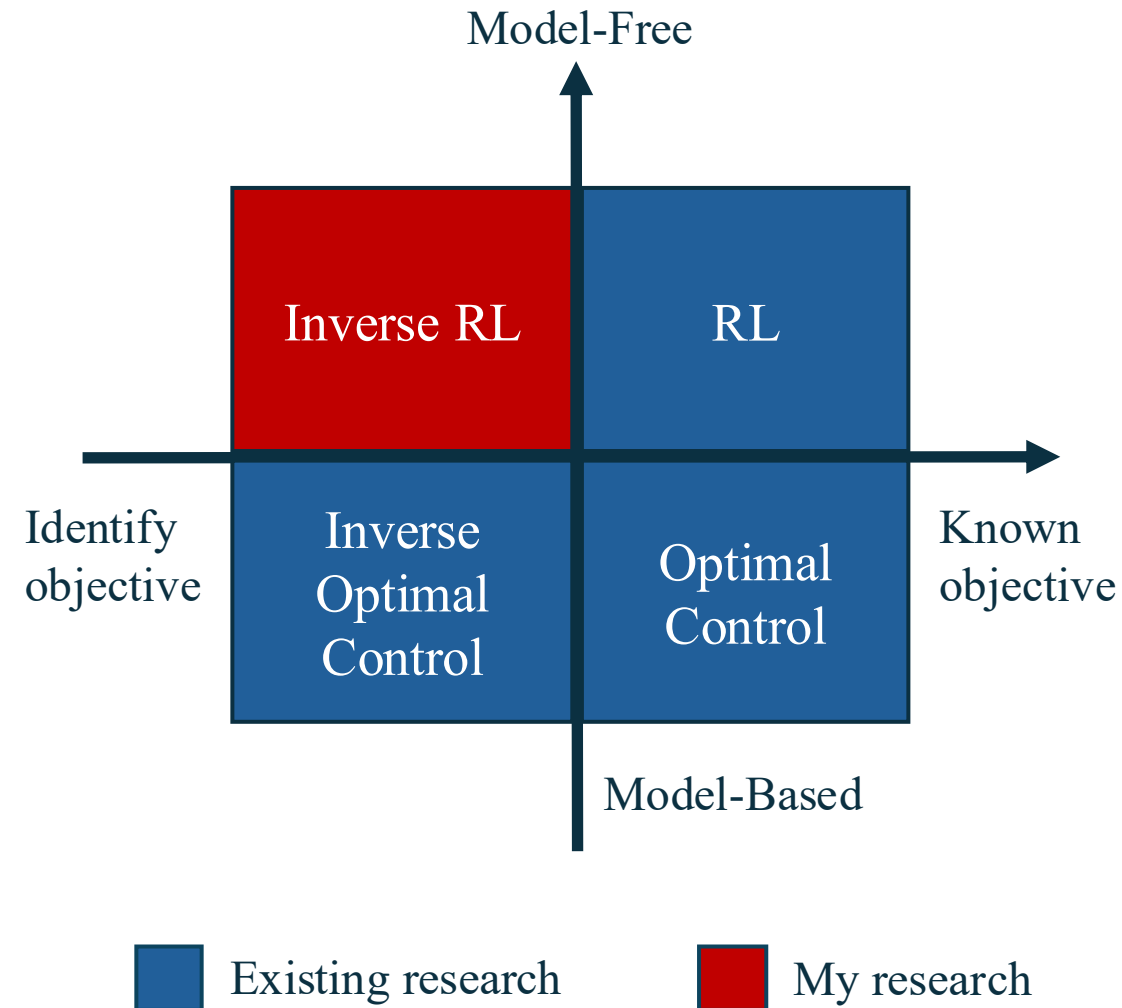
Trajectories of frequency, generator output, governor valve, and tie-line power of i -th area system



Convergence of trained parameters to the desired ones

Inverse Reinforcement Learning

- A new discipline of inverse RL in control theory



Imitation Control with Inverse Reinforcement Learning

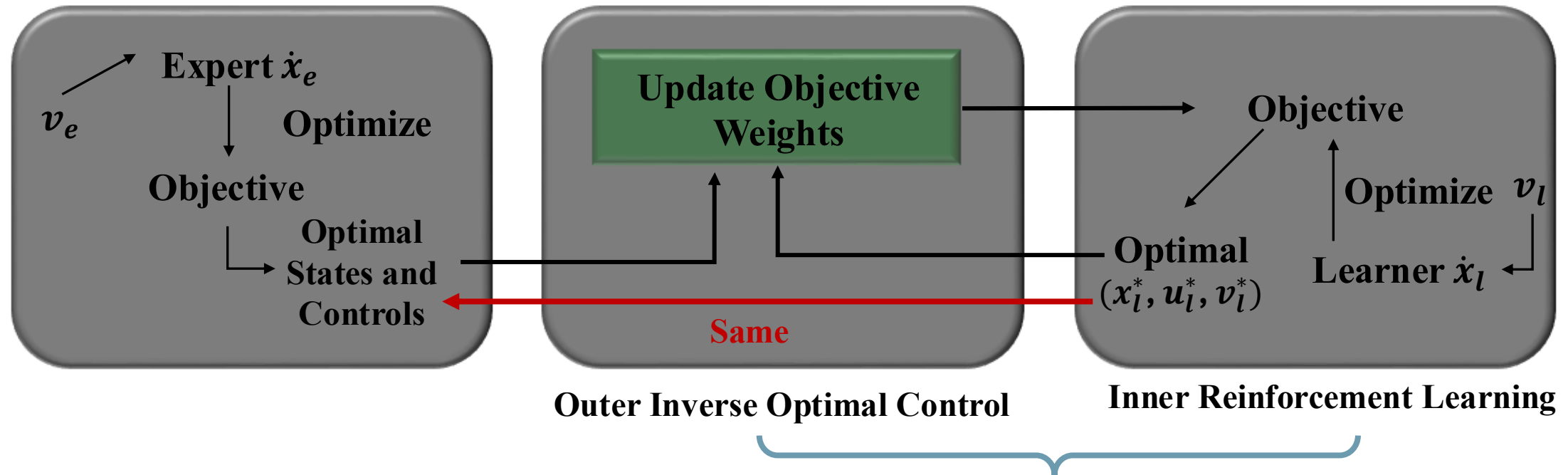
Expert

Dynamics: $\dot{x}_e = f(x_e) + g(x_e) u_e + h(x_e) v_e$
 Objective: $V_e = \int_0^\infty x_e^T Q_e x_e + u_e^T R_e u_e - v_e^T S_e v_e d\tau$

Learner

Dynamics: $\dot{x}_l = f(x_l) + g(x_l) u_l + h(x_l) v_l$
 Trail Objective: $V_l = \int_0^\infty x_l^T Q_l x_l + u_l^T R_l u_l - v_l^T S_l v_l d\tau$

UNKNOWN



Two-loop inverse Reinforcement Learning

➤ Model-Free Data-Driven Inverse Reinforcement Learning Algorithm

Algorithm 1 Model-free Integral Inverse RL Algorithm for Adversarial Apprentice Game

1: **Initialization:** select $Q_l^0 \geq 0$, $R_l > 0$, $S_l > 0$, initial stabilizing u_l^{00} , and small thresholds ε_1 , ε_2 . Set $j = 0$. Apply stabilizing u_l to the learner dynamics (27).

2: **Outer j iteration loop based on inverse optimal control**

3: **Inner i iteration loop using optimal control:** given j , set $i = 0$, and use initial stabilizing u_l^{j0} .

4: Off-policy Integral RL for solving V_l^j , u_l^j and v_l^j

$$\begin{aligned} & V_l^{ji}(x_l(t+T)) - V_l^{ji}(x_l(t)) \\ & - \int_t^{t+T} 2 \left((u_l^{j(i+1)})^T R_l (u_l - u_l^{ji}) \right. \\ & \quad \left. - (v_l^{j(i+1)})^T S_l (v_l - v_l^{ji}) \right) d\tau \\ & = \int_t^{t+T} \left(-q^T(x_l) Q_l^j q(x_l) \right. \\ & \quad \left. - (u_l^{ji})^T R_l u_l^{ji} + (v_l^{ji})^T S_l v_l^{ji} \right) d\tau. \end{aligned}$$

5: Stop if $\|V_l^{ji} - V_l^{j(i-1)}\| \leq \varepsilon_1$, then set $V_l^j = V_l^{ji}$, $u_l^j = u_l^{ji}$ and $v_l^j = v_l^{ji}$. Otherwise, set $i \leftarrow i + 1$ and go to Step 4.

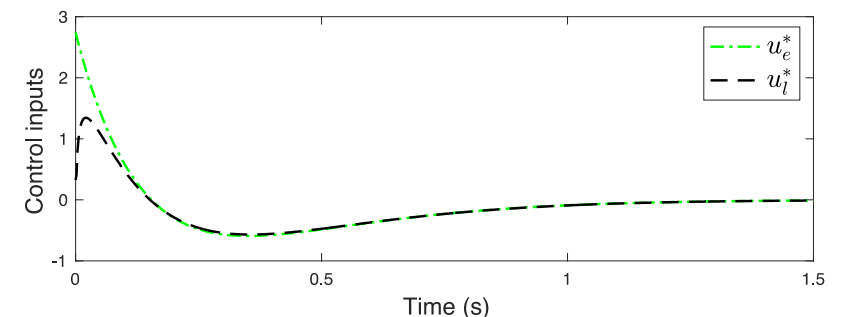
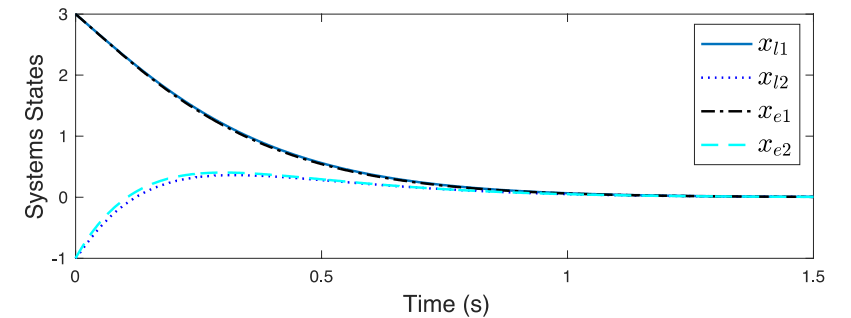
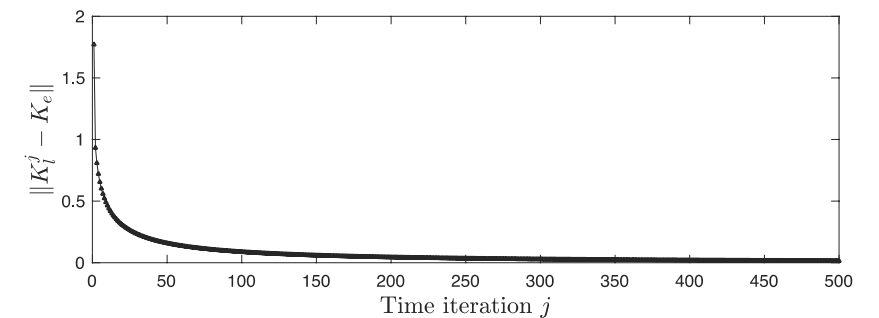
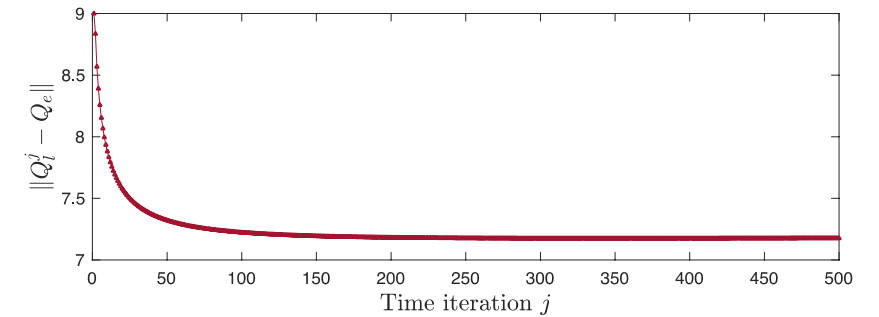
6: **State-penalty weight Q_l^{j+1} update using the expert's demonstration u_e^***

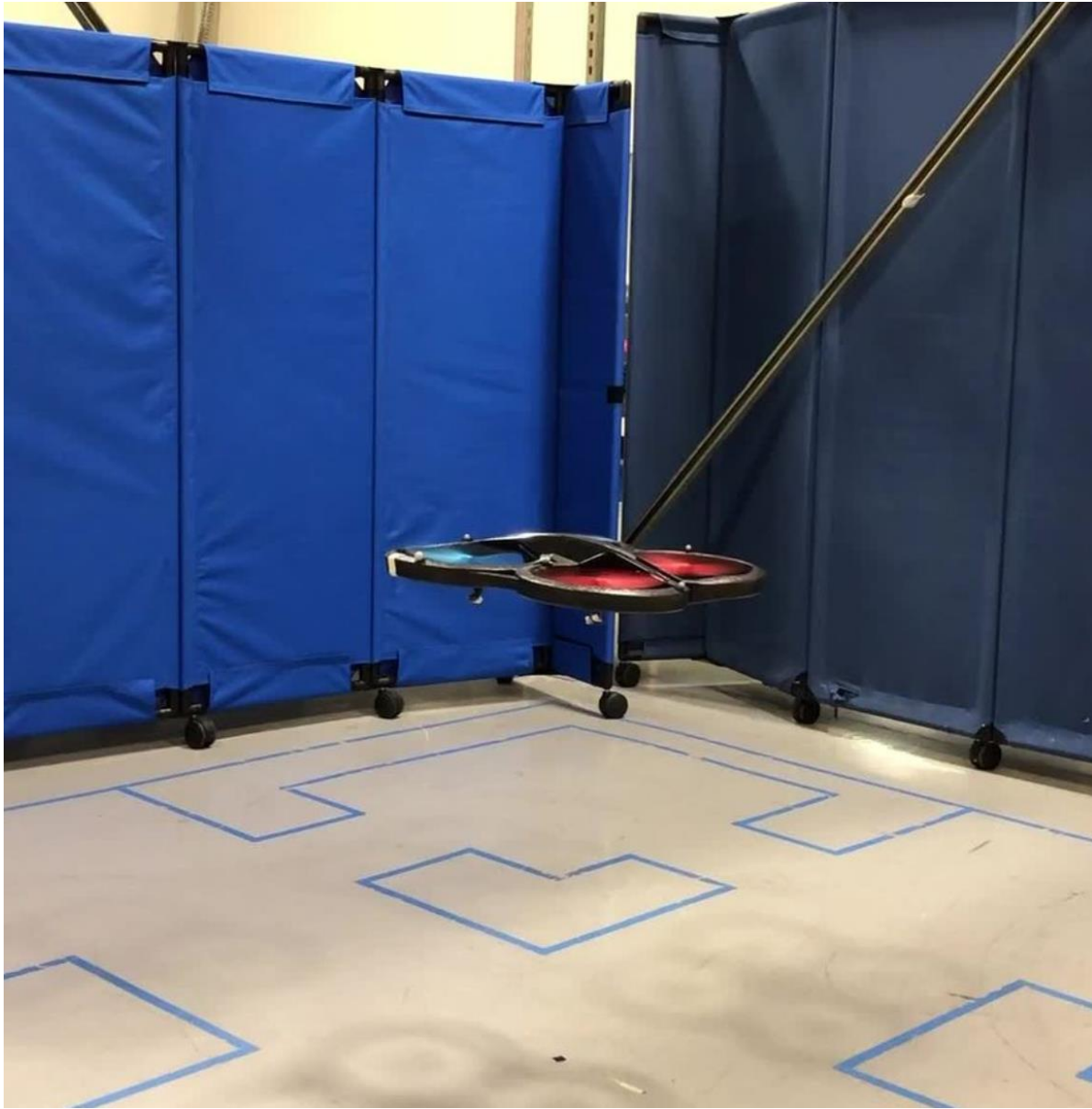
$$\begin{aligned} & \int_t^{t+T} q^T(x_l) Q_l^{j+1} q(x_l) d\tau \\ & = \int_t^{t+T} \left(u_e^{*T} R_l u_e^* - 2u_e^{*T} R_l u_l^j + (v_l^j)^T S_l v_l^j \right) d\tau \\ & \quad - V_l^j(x_l(t+T)) + V_l^j(x_l(t)). \end{aligned}$$

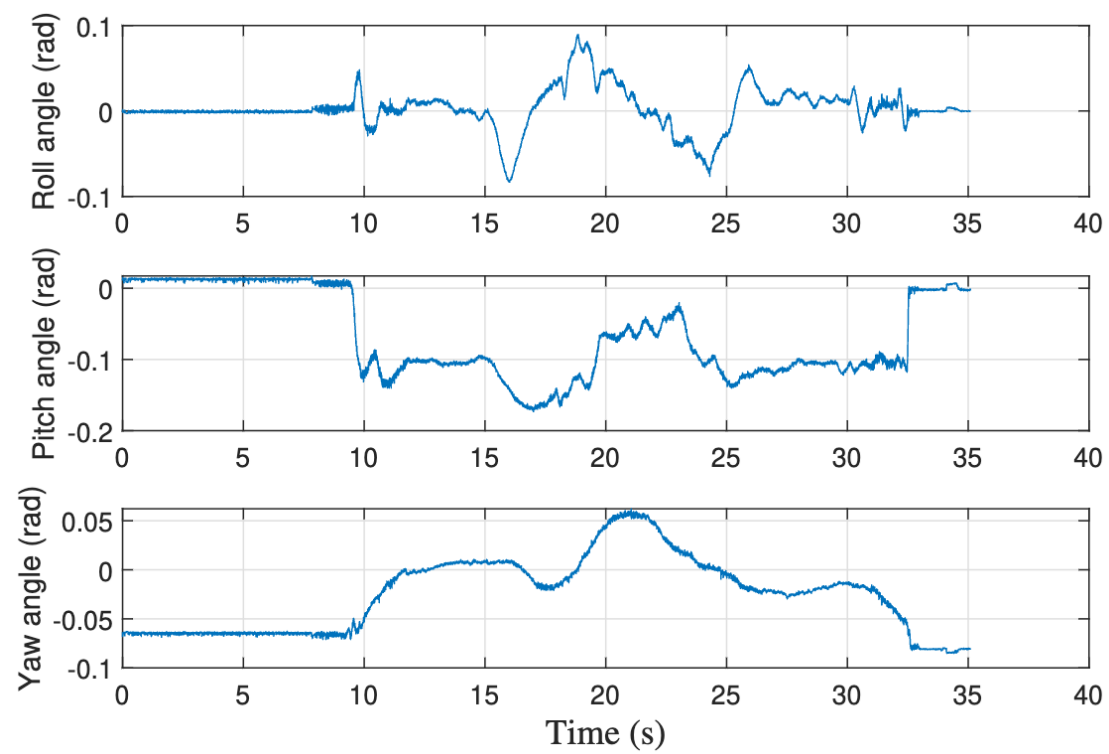
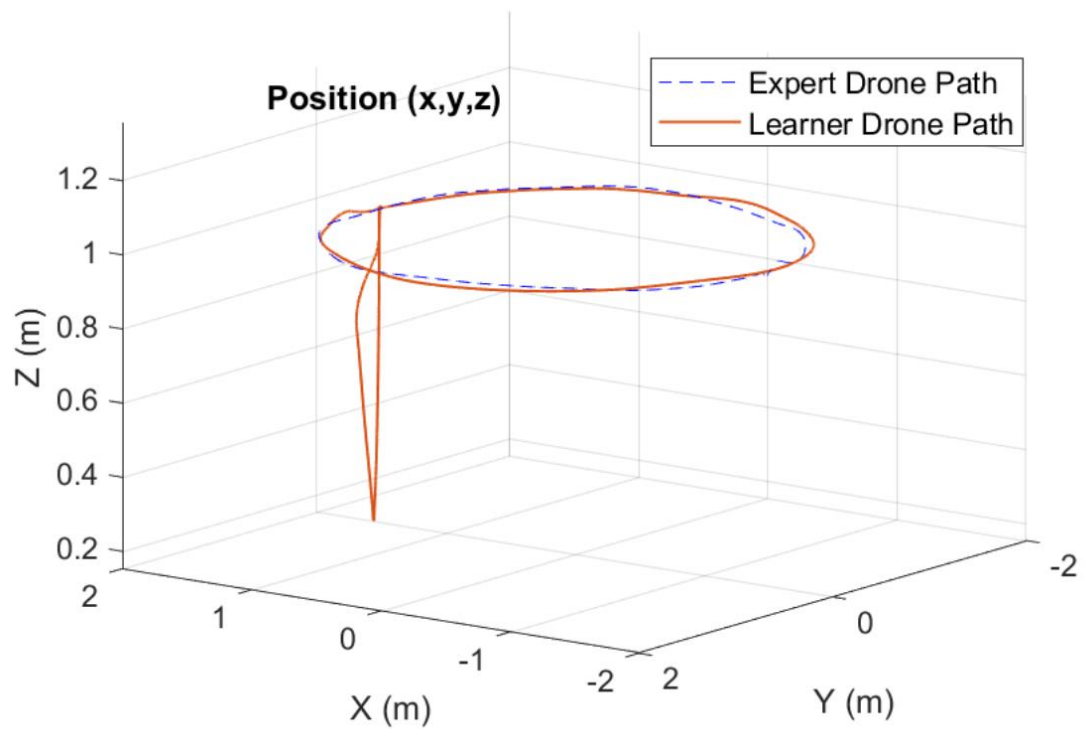
7: **Stop if $\|Q_l^{j+1} - Q_l^j\| \leq \varepsilon_2$.** Otherwise, set $u_l^{(j+1)0} = u_l^j$ and $j \leftarrow j + 1$, then go to Step 3.

Reinforcement Learning
(Off-Policy)

Inverse Optimal
Control







Distributed Kalman Consensus Filter for Estimation with Moving Targets

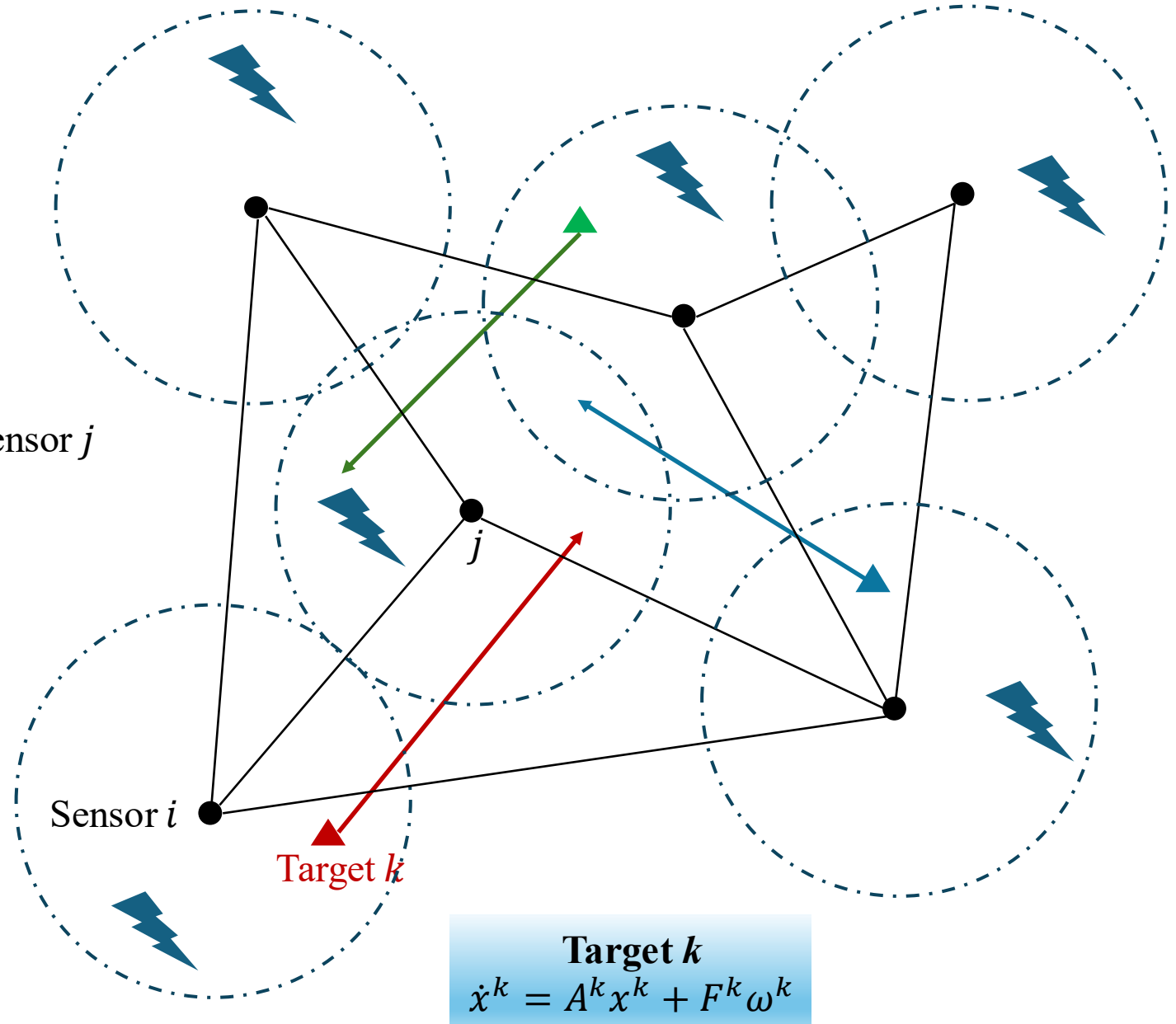


Direct

Indirect

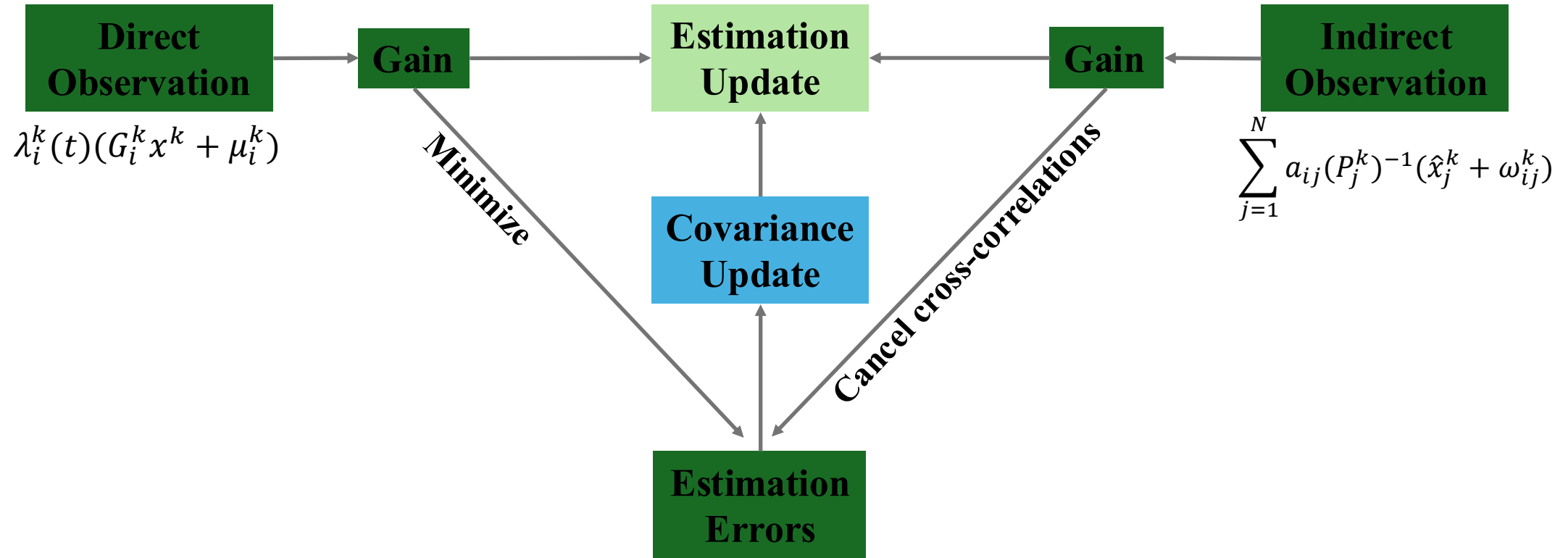
Sensor i 's measurement of target k

$$z_i^k = \begin{bmatrix} \lambda_i^k(d_i^k, t)(G_i^k x^k + \mu_i^k) \\ \sum_{j=1}^N a_{ij}(P_j^k)^{-1}(\hat{x}_j^k + \omega_{ij}^k) \end{bmatrix}$$



Target k

$$\dot{x}^k = A^k x^k + F^k \omega^k$$



Distributed Estimation Update

$$\begin{aligned} \dot{\hat{x}}_i^k = & A^k \hat{x}_i^k - \lambda_i^k P_i^k (G_i^k)^T (R_i^k)^{-1} (G_i^k (x^k - \hat{x}_i^k) + \mu_i^k) \\ & + 2d_i^k P_i^k \left[\sum_{j=1}^N a_{ij}^k (P_j^k)^{-1} (P_i^k + P_j^k) \right]^{-1} \sum_{j=1}^N a_{ij}^k (P_j^k)^{-1} (\hat{x}_j^k - \hat{x}_i^k) \end{aligned}$$

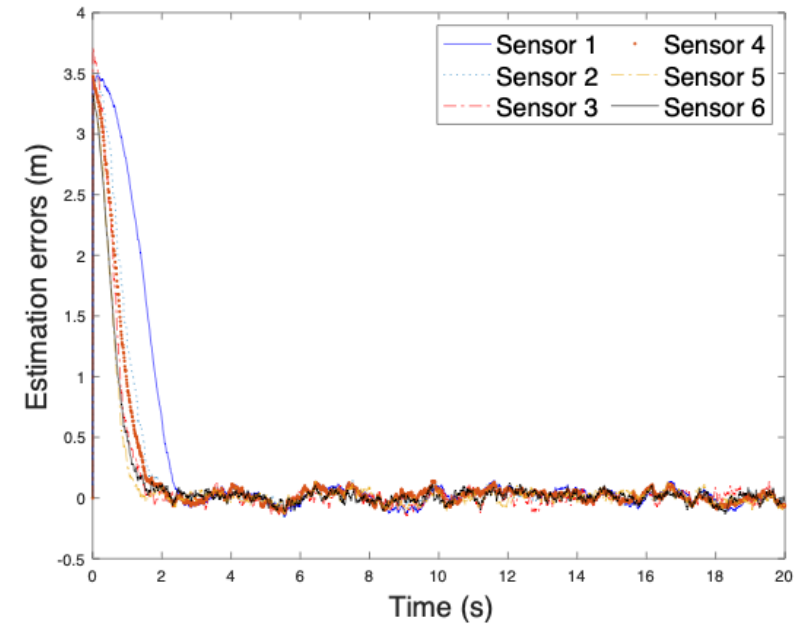
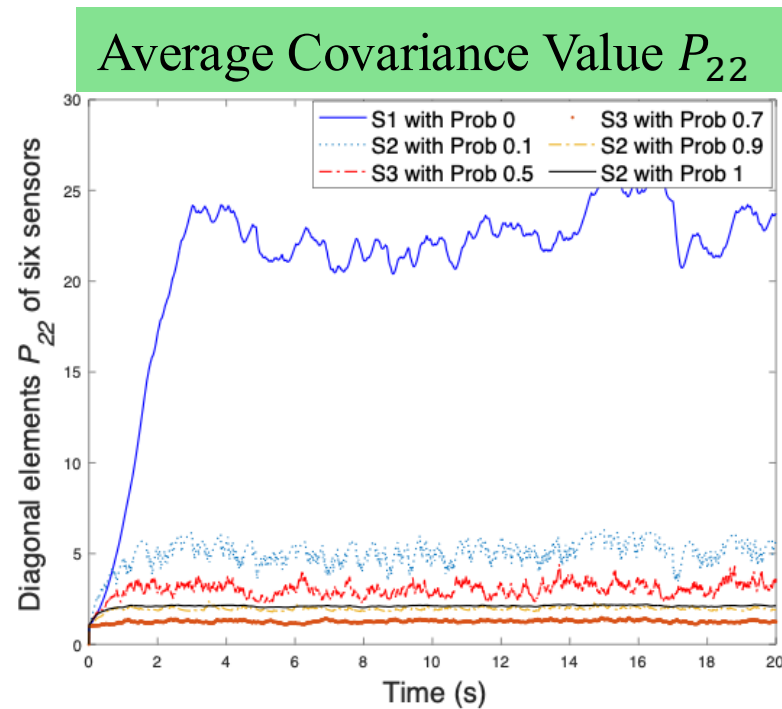
Distributed Covariance Update

$$\begin{aligned} \dot{P}_i^k = & A_i^k P_i^k + P_i^k (A_i^k)^T - \lambda_i^k P_i^k (G_i^k)^T (R_i^k)^{-1} G_i^k P_i^k + F^k W^k (F^k)^T \\ & - 4 \sum_{j=1}^N a_{ij}^k P_i^k \left[\sum_{j=1}^N a_{ij}^k (P_j^k)^{-1} (P_i^k + P_j^k) \right]^{-1} \sum_{j=1}^N a_{ij}^k (P_j^k)^{-1} P_i^k \end{aligned}$$

Effectiveness

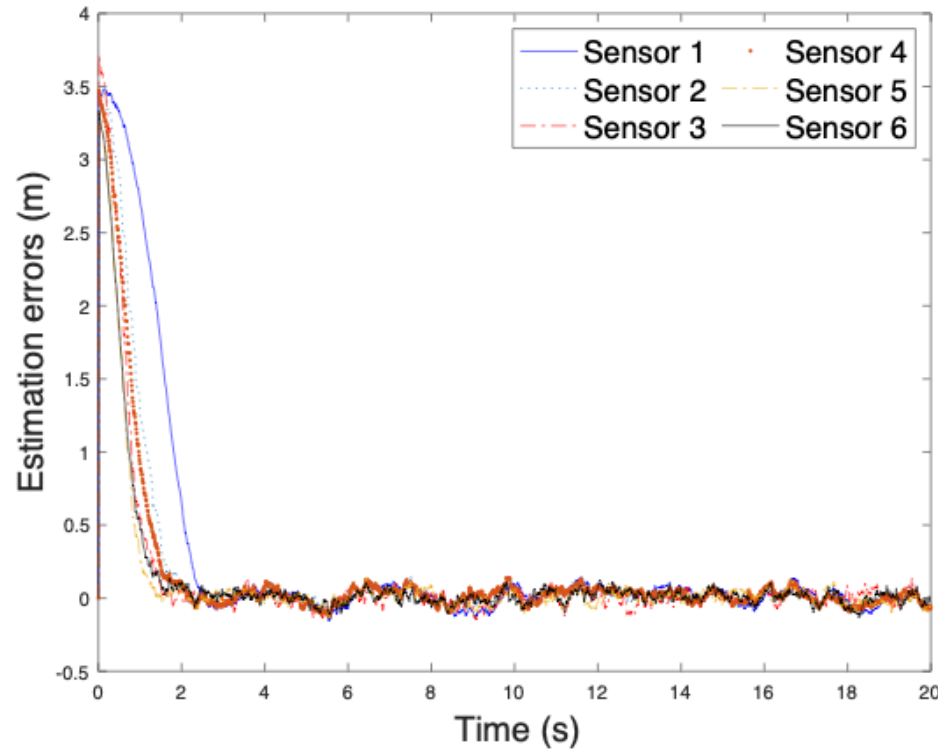


Fig. 1 Ford Taurus

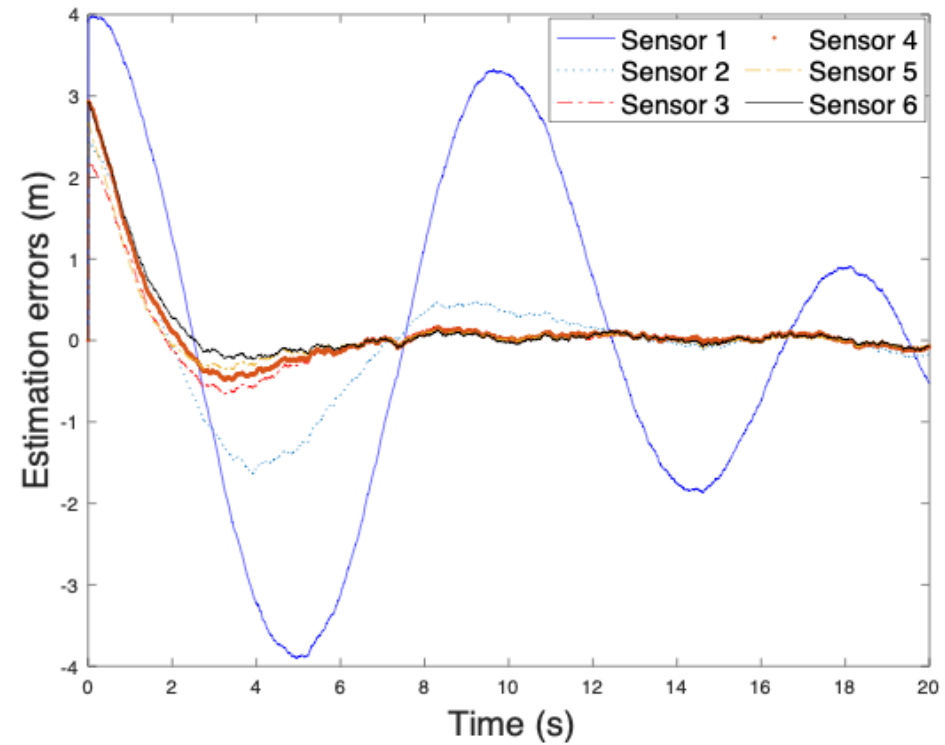


Faster Convergence

The Convergence of Average Estimation Errors of Each Sensor



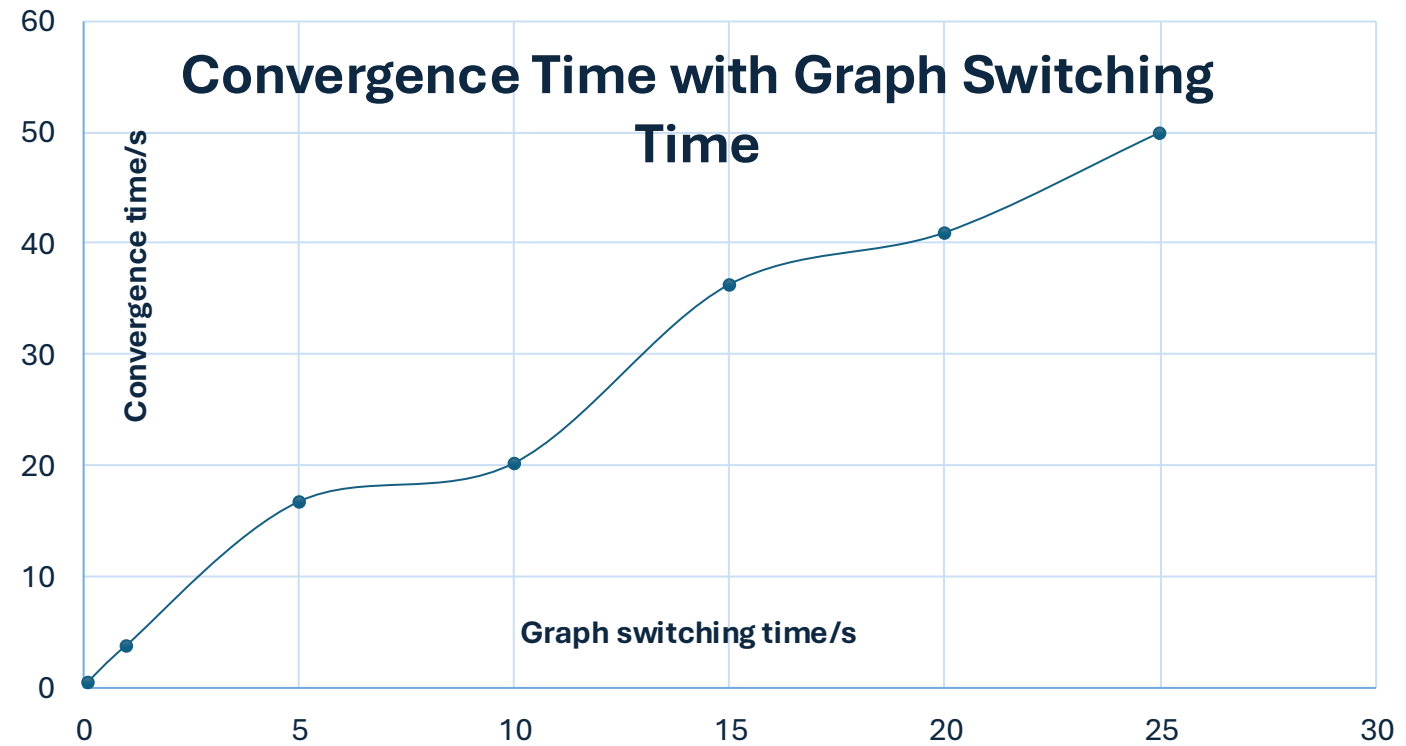
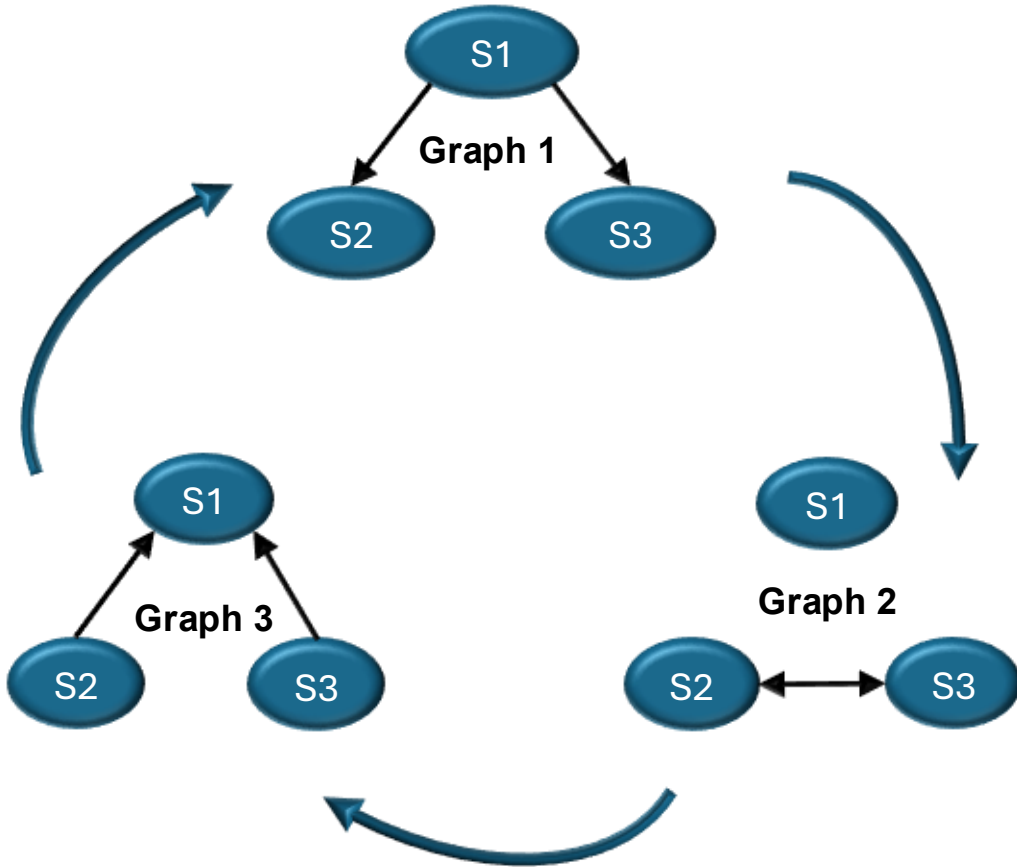
By DKCF



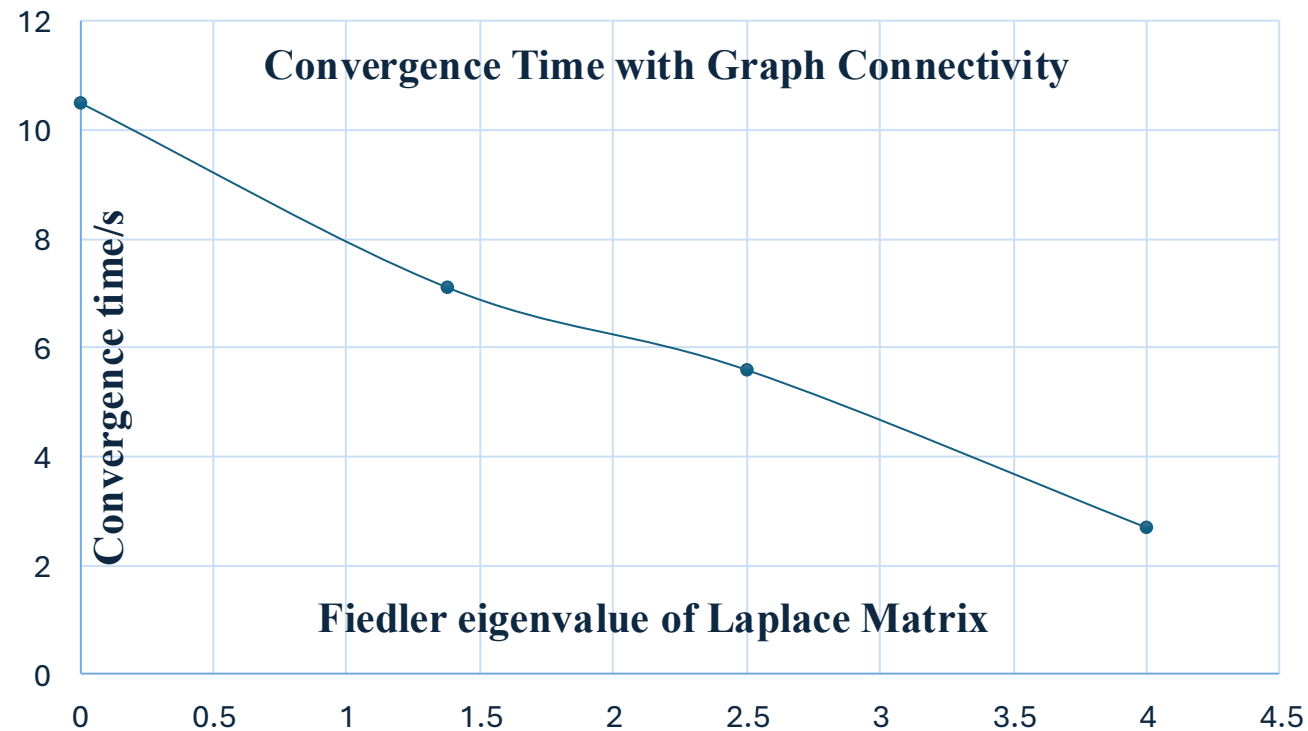
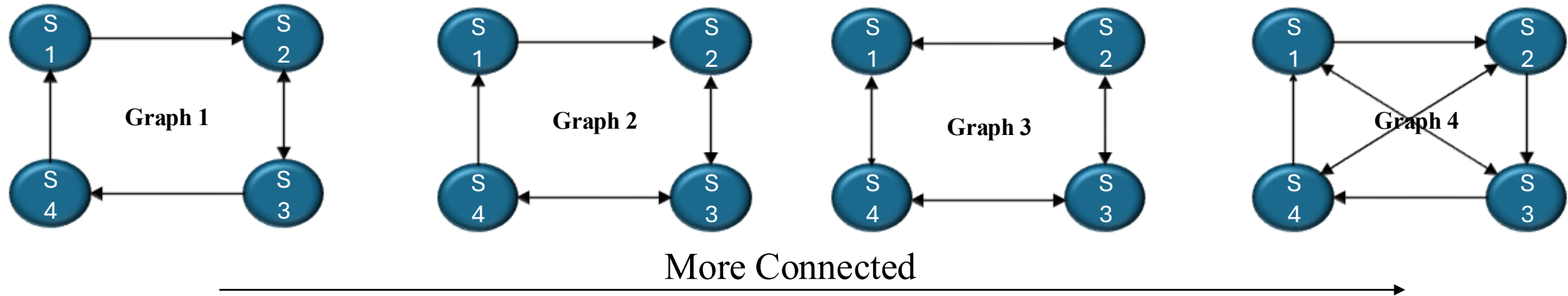
By KCF^[5]

[5] Olfati-Saber, Reza. "Distributed Kalman filtering for sensor networks." *2007 46th IEEE Conference on Decision and Control*.

Convergence with Graph Switching



Convergence with Graph Connectivity



Thank you!